

Project Documentation

HR Analytics — Job Change Prediction

1. Introduction

1.1 Project Background

Organizations need effective recruitment and candidate screening strategies to identify candidates who are more likely to seek new job opportunities. Machine Learning can help analyze candidate information and identify patterns associated with job-change intentions.

1.2 Problem Statement

The objective of this project is to build a Machine Learning-based candidate screening system that predicts whether a candidate is likely to be looking for a new job.

1.3 Project Objective

The main objectives are:

- Prepare and clean the recruitment dataset.
 - Apply feature engineering and data transformation.
 - Build two Machine Learning models.
 - Compare their performance using multiple evaluation metrics.
 - Identify important factors associated with job-change predictions.
 - Generate recruitment insights and recommendations.
 - Develop a recruitment dashboard.
 - Provide a candidate ranking system as an optional bonus.
-

2. Dataset Description

The project uses the **HR Analytics Job Change of Data Scientists** dataset.

The dataset contains information about candidates, including their education, experience, company information, training hours, and relevant work experience.

Main Features Used

Feature	Description
Education Level	Candidate's education level
Experience	Years of professional experience
Company Type	Type of company where the candidate works

Feature	Description
Training Hours	Number of training hours completed
Relevant Experience	Whether the candidate has relevant experience
Target	Whether the candidate is looking for a new job

The target variable is binary:

- 0 → Candidate is not looking for a new job
- 1 → Candidate is looking for a new job

3. Data Preparation

3.1 Missing Value Handling

Missing values were identified in several categorical features, including education level, company type, and relevant candidate information.

Missing values were handled before applying Machine Learning algorithms.

For the experience feature, the Unknown values were converted into missing values and replaced using the median after numerical transformation.

3.2 Feature Engineering

The original experience feature contained categorical values such as <1, numerical years, and >20.

A new numerical feature called:

experience_years

was created.

The transformation included:

- <1 → 0.5
- 1–20 → corresponding numerical value
- >20 → 21
- Unknown → median value

This allowed the experience information to be represented numerically and used effectively by the Machine Learning models.

3.3 Data Transformation

Categorical variables were converted into numerical variables using **One-Hot Encoding**.

This transformation was required because Machine Learning models work with numerical inputs.

After preprocessing and feature engineering, the dataset was divided into:

- **Training set:** 15,326 records
- **Testing set:** 3,832 records

A random_state of 42 was used to make the split reproducible.

4. Machine Learning Models

Two classification models were developed according to the project requirements.

4.1 Logistic Regression

Logistic Regression was used as a baseline classification model for predicting whether a candidate belongs to target = 0 or target = 1.

4.2 Random Forest

Random Forest was used as a second classification model. It combines multiple decision trees and can capture more complex relationships between candidate features and the target variable.

5. Model Evaluation

The models were evaluated using the required metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Performance Comparison

Metric	Logistic Regression	Random Forest
Accuracy	76%	70%
Precision	56%	39%
Recall	20%	33%
F1 Score	30%	36%

Interpretation

Logistic Regression achieved higher **Accuracy** and **Precision**.

However, Random Forest achieved higher **Recall** and **F1 Score** for target = 1.

Since the project's screening objective is to identify more candidates who are likely to change jobs, **Random Forest is the preferred model when recall is the main priority.**

6. Business Intelligence

6.1 Most Important Hiring Factors

The Random Forest feature importance analysis identified the following as the most important features:

Feature	Importance
Training Hours	67.60%
Experience Years	18.56%
Company Type — Unknown	3.62%
Education — High School	1.77%
Company Type — Pvt Ltd	1.70%

Insight

Training Hours was the most important feature in the Random Forest model, followed by Experience Years.

Important: Feature importance indicates how useful a feature was to the model's predictions; it does not prove that the feature directly causes job changes.

6.2 Candidate Profile Trends

The job-change rate was analyzed by education level.

Education Level Job Change Rate

Graduate	27.98%
Unknown	22.61%
Masters	21.44%
High School	19.53%
PhD	14.01%
Primary School	13.31%

Insight

Candidates with a **Graduate education level** had the highest job-change rate at **approximately 28%**.

6.3 Recruitment Statistics

The dataset was also analyzed to identify overall recruitment statistics:

- Total number of candidates
- Number of candidates with target = 1
- Overall job-change rate

These statistics are included in the Recruitment Dashboard to provide a quick overview of the dataset.

7. Recruitment Dashboard

The dashboard presents the main project findings in a visual format.

It includes:

Recruitment Statistics

- Total Candidates
- Candidates likely to change jobs
- Job Change Rate

Model Performance

- Accuracy
- Precision
- Recall
- F1 Score

Important Hiring Factors

- Training Hours
- Experience Years
- Other important features

Candidate Profile Trends

- Job-change rate by education level

The dashboard provides recruiters with a concise view of the model results and recruitment patterns.

8. Hiring Recommendations

Based on the analysis, the following recommendations were identified:

1. **Focus on training and development opportunities**, as training hours were the most important feature in the model.
 2. **Consider professional experience during candidate screening**, as experience years was the second most important feature.
 3. **Use Random Forest when identifying potential job changers is the main priority**, because it achieved higher Recall and F1 Score for the positive class.
 4. **Use model predictions as decision-support information**, rather than making recruitment decisions based only on the model.
-

9. Optional Bonus — Candidate Ranking

As an optional bonus, a candidate ranking system was developed using the Random Forest predicted probabilities.

The system calculates the probability that each candidate belongs to target = 1 and ranks candidates from highest to lowest probability.

The **Top 10 candidates** were then selected based on their predicted probability.

This ranking can help recruiters prioritize candidates for further review.

10. Conclusion

This project developed a Machine Learning-based candidate screening system using Logistic Regression and Random Forest.

The data was cleaned, transformed, and enhanced through feature engineering before model training.

The evaluation showed that Logistic Regression achieved higher overall Accuracy and Precision, while Random Forest achieved higher Recall and F1 Score for potential job changers.

Based on the project's objective of identifying more potential job changers, **Random Forest was selected as the preferred screening model**.

The project also provided business insights into important hiring factors, candidate profiles, and recruitment statistics, supported by a recruitment dashboard and an optional candidate ranking system.